

LifeBudget Micro

User Guide / User Manual

Final Submission Version - module reference, quick-start guide, common interface notes, and risk/universe, strategy engine, long-term scenario, and final report guidance

Educational personal finance and investment scenario-planning prototype

Purpose of this guide

This document explains how to run and use LifeBudget Micro, what each module does, what inputs are expected, what options are available to the user, what outputs are produced, and which limitations should be considered when interpreting the results.

Important educational notice

LifeBudget Micro is not financial advice, investment advice, or a personal suitability assessment. Investment outputs are historical backtests or educational proxies and do not guarantee future outcomes.

How this guide is organised

The guide combines a short quick-start flow for markers with a module-by-module user manual. It does not list every possible interface state, but it documents the main user-relevant setup routes, controls, outputs, and interpretation points.

1. Project Overview

LifeBudget Micro is an educational Streamlit prototype for personal finance planning and investment scenario exploration. It helps users estimate savings capacity, build an educational investment strategy, run historical backtests, and compare long-term savings-only versus savings-plus-investing scenarios.

The system is designed to demonstrate a working end-to-end software artefact: user financial inputs are converted into budget feasibility outputs, connected to an investment strategy engine, and then used in long-term scenario projections.

2. Core Features Implemented

- Personal finance planning: weekly income and spending inputs, savings capacity estimation, and short-term target feasibility checks.
- Risk profile and asset universe setup: user-selected risk profile, asset universe selection, and market panel preparation.
- Strategy engine: walk-forward historical backtesting with risk/return metrics and benchmark context.
- Strategy improvement checks: optional suggestions and diagnostics designed to compare alternative setups or assess robustness.
- Reliability and robustness checks: educational diagnostics that help users understand whether a strategy result appears stable or sensitive to assumptions.
- Long-term scenario projection: Monte Carlo-style comparison of savings-only and savings-plus-investing outcomes.
- Final report / insights: a summary view that connects personal finance inputs, investment strategy outputs, projections, and limitations.

3. Setup and Run Instructions

3.1 Running the Online Version

The app can be accessed through the hosted Streamlit deployment:

```
https://lifebudget-micro.streamlit.app/
```

This is the recommended way to test the submitted prototype quickly. The online version may take longer during market panel generation or strategy backtesting because it runs on hosted Streamlit resources.

3.2 Device Compatibility

LifeBudget Micro is a web-based Streamlit app and can be opened from a browser on a desktop, laptop, tablet, or mobile device.

The recommended testing environment is a desktop or laptop browser. The app includes charts, tables, expandable diagnostics, and sidebar navigation, which are easier to review on a larger screen.

Mobile browser access may work for quick inspection, but the prototype has primarily been tested and optimised for desktop/laptop use.

3.3 Repository and Final Submission Snapshot

The submitted GitHub repository should represent a markable snapshot of the final artefact.

The hosted Streamlit app is deployed from the `deploy-lifebudget-demo` branch. The final assessment tag `assignment-3-final-v4` should point to the same deployed commit, so the marker can use the online app while also inspecting the exact code snapshot submitted for assessment.

GitHub repository: <https://github.com/wirstyle/lifebudget-micro>

Hosted app	https://lifebudget-micro.streamlit.app/
Deployed branch	deploy-lifebudget-demo
Final assessment tag	assignment-3-final-v4

For clarity, the final commit should be labelled with a clear Git tag or GitHub release. A suggested tag name is:

assignment-3-final-v4

If a different tag or release name is used, replace the suggested label above with the exact submitted version label so the marker can identify the final assessment snapshot.

3.4 Running Locally

Clone the repository and install the required dependencies:

```
git clone https://github.com/wirstyle/lifebudget-micro.git
cd lifebudget-micro
pip install -r requirements.txt
```

Run the Streamlit app:

```
streamlit run app.py
```

The app should then open in the browser. If it does not open automatically, copy the local Streamlit URL shown in the terminal into the browser.

3.5 Test Credentials and Sample Data

No login credentials are required. The app does not require the user to upload private financial data. Users can enter sample budget values directly in the interface. Market data is prepared through the built-in asset panel workflow.

Suggested Personal Finance sample values

Take-home income: £460/week | Fixed essentials: £185/week | Variable essentials: £80/week | Discretionary spending: £85/week | Approximate free margin: £110/week | Example savings target: £32/week | Example horizon: 52 weeks

4. How to Use This Guide

This document is a user guide rather than a pure tutorial. The quick-start section gives markers a short path for testing the full system, while the module reference sections explain what each part of the app does, what the user can set, what outputs are produced, and how to interpret those outputs.

The guide focuses on user-relevant choices and main configuration routes. It does not document every possible UI state or every internal diagnostic value, because many diagnostics are optional and are mainly included for transparency and evaluation.

5. Quick Start Flow for Markers

1. Open the app online or run it locally.
2. Accept the educational notice on the home screen.
3. Open Personal Finance Planner.
4. Enter or keep sample weekly income, spending, and savings target values.

5. Review the cash-flow summary, free margin, savings target, feasibility check, and short-term path chart.
6. Continue to Investment Strategy Lab.
7. Select a risk profile and prepare the asset universe / market panel.
8. Run the Strategy Engine.
9. Review historical risk/return metrics, benchmark context, reliability diagnostics, and optional improvement checks.
10. Continue to Long-Term Scenario Explorer.
11. Generate a long-term projection using the monthly contribution produced by the finance setup.
12. Open the final report / insights summary and review the connected explanation.

This flow demonstrates the main idea of the project: a user can move from personal finance inputs to investment strategy testing and then to long-term educational scenario comparison.

6. Common Interface Elements

LifeBudget Micro uses a consistent interface pattern across the app. This section explains the elements that appear repeatedly so the user understands how to navigate the prototype before reading the individual module sections.

6.1 Sidebar Navigation

The sidebar remains visible throughout the system. It is used as a navigation and guidance area rather than only as a menu.

- Home button: returns the user to the main module selection screen. On the home screen itself, this button is disabled because the user is already there.
- Current module context: shows the active module name and a short description of what the user is doing.
- Current state summary: displays relevant saved information, such as the finance plan, risk profile, prepared asset universe, or latest strategy result.
- Contextual guidance: explains what the user should check before continuing to the next module.
- Help expanders: provide Q&A, terms, assumptions, limitations, or diagnostics depending on the module.

6.2 Expanders and Progressive Disclosure

Expandable sections are used to keep the interface readable. Main actions are shown directly on the page, while supporting explanations and technical details are placed inside expanders.

Expander type	Purpose
Q&A sections	Answer common user questions in plain language, such as why weekly values are used or what a feasibility check means.
Terms sections	Define important terms used in the current module, such as free margin, drawdown, volatility, Sharpe ratio, or Monte Carlo projection.
Diagnostics sections	Show technical or audit-style information, such as market panel status, timing information, reliability checks, or engine details.
Assumptions and limitations	Explain caveats behind educational simulations, backtests, projections, and non-advice disclaimers.
Detailed controls	Allow users to make more precise edits only when needed, without making the main workflow too complex.

6.3 Status and Information Messages

The app uses colour-coded messages as part of the user guidance system. These messages should be read as contextual help rather than financial recommendations.

- Blue information boxes explain the purpose of a section, the recommended next action, or how a result should be interpreted.
- Green confirmation boxes indicate that an action has been completed successfully, such as accepting the educational notice, saving a plan, or preparing a panel.
- Yellow warning boxes highlight assumptions, educational caveats, limitations, or non-advice disclaimers.

6.4 Module Cards and Recommended Flow

The app is organised around three main modules: Personal Finance Planner, Investment Strategy Lab, and Long-Term Scenario Explorer. Users can inspect modules directly, but the recommended workflow gives the clearest end-to-end demonstration because later modules reuse outputs created earlier in the process.

Recommended full workflow

Finance Planner -> Investment Lab -> Scenario Explorer -> Insights Summary

7. Home and Module Selection

When the app first opens, the user is shown the main home screen. This page introduces LifeBudget Micro as an educational planning and scenario-comparison prototype.

Before using the modules, the user must accept the educational notice. This is required because the app produces educational backtests, scenario simulations, and projection outputs rather than financial advice or guaranteed predictions.

7.1 User Choices on This Screen

- Review the educational assumptions and limitations.
- Accept the educational notice to enable the module cards.
- Open one of the three main modules: Personal Finance Planner, Investment Strategy Lab, or Long-Term Scenario Explorer.
- Use the sidebar Home button later to return to this screen.

7.2 Module Cards

- Personal Finance Planner: builds the budget baseline, estimates savings capacity, and tests short-term target feasibility.
- Investment Strategy Lab: prepares an investment universe, runs the strategy engine, and reviews risk/return diagnostics.
- Long-Term Scenario Explorer: compares savings-only, educational proxy, and tested-strategy long-term scenarios.

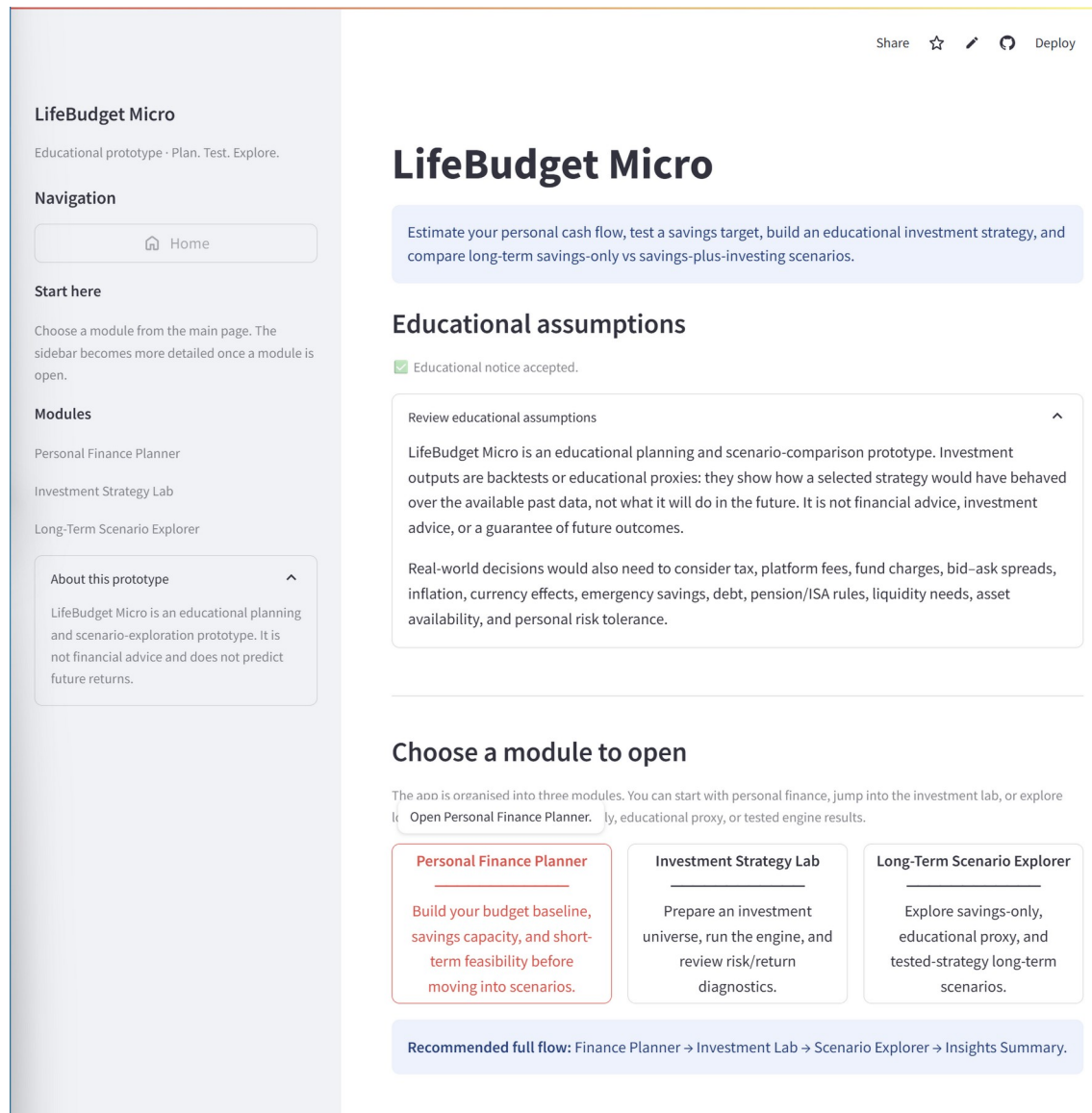


Figure 1. Home screen and module selection after accepting the educational assumptions.

8. Personal Finance Planner

The Personal Finance Planner is the first recommended module in the full workflow. It creates a simple weekly budget baseline, estimates the user's free margin, and turns that free margin into a savings target that can later be used by the investment and long-term scenario modules.

This module is intentionally designed around approximate weekly values. The aim is not to recreate a complete bank statement, but to create a clear planning baseline that can be used consistently across the rest of the prototype.

8.1 Purpose of the Module

- Estimate weekly income, spending, and free margin.
- Create a short-term savings target from the available free margin.
- Check whether the selected target appears feasible under uncertainty.
- Provide a contribution bridge for later investment and long-term scenario modules.

8.2 Ways to Set the Budget

Route	What the user does	When it is useful
Quick presets	Choose Quiet, Typical, or Expensive.	Fastest route for a demo or rough planning baseline.
Budget sliders	Adjust weekly income, fixed essentials, variable essentials, and discretionary spending.	Best route for simple manual tuning.
Gross-pay estimate	Enter gross income, income period, and estimated deduction rate.	Useful when the user knows gross income but not take-home pay.
Fixed essentials breakdown	Itemise rent/housing, utilities, council tax, internet/phone, transport pass, insurance, and subscriptions.	Useful when fixed commitments need to be transparent.
Variable essentials breakdown	Estimate groceries, utilities, household basics, and commute costs.	Useful when variable needs should be broken down.
Exact totals editor	Enter income and spending totals directly with their periods.	Useful when the user already knows the exact totals they want to test.

8.3 Core Inputs

- Take-home income: money available before weekly spending is subtracted.
- Fixed essentials: harder-to-change commitments such as rent, bills, and subscriptions.
- Variable essentials: necessary but variable costs such as groceries, utilities, and commute basics.
- Discretionary spending: flexible lifestyle spending.
- Savings target: weekly amount tested against the current free margin.
- Short-term horizon: period over which the target path is checked.

8.4 Optional Tools

The optional tools are not required for the basic demo flow. They all update the same underlying budget baseline, so the user can choose whichever route is most convenient.

- Estimate take-home income from gross pay: estimates take-home income from a gross income value, income period, and deduction rate.
- Build fixed essentials breakdown: creates a weekly fixed-essentials total from itemised recurring costs.
- Build variable essentials breakdown: creates a weekly variable-essentials total from groceries, utilities, household basics, and commuting assumptions.
- Edit exact income and spending totals: allows direct entry of the main totals and their periods.

8.5 Main Outputs

- Budget summary: weekly, monthly, and yearly view of income, essentials, discretionary spending, and free margin.
- Cash-flow summary: compact view of income, spending, and free margin.
- Savings target card: lets the user test cautious, balanced, or stretch savings habits.
- Feasibility check: shows conservative, expected, and high-case outcomes for the short-term plan.
- Short-term path chart: compares the baseline path with the savings-target path over the selected horizon.
- Uncertainty band: shows the 10-90% range around the target plan when short-term variability is applied.
- Sidebar current plan: keeps the key values visible while the user edits the module.

8.6 Example Test Values

Example used for testing

Income: £460/week | Spending: £350/week | Free margin: £110/week | Savings target: £32/week | Target uses about 29% of free margin | Horizon: 52 weeks

8.7 How to Interpret the Outputs

A positive free margin suggests that the entered assumptions leave potential savings capacity. A negative or very small free margin suggests that the plan may need adjustment before testing longer-term scenarios.

The feasibility check is not a guarantee. It is a short-term planning stress check based on the values entered into the prototype. Its purpose is to show whether the selected savings target still appears workable under uncertainty.

The key output for the rest of the app is the savings/contribution bridge. This value connects the Personal Finance Planner to the Investment Strategy Lab and the Long-Term Scenario Explorer.

8.8 Caveats for This Module

- The figures are planning estimates, not verified bank data.
- The module does not assess whether the user's real financial situation is suitable.
- Weekly values are used for consistency and simplicity across the prototype.
- Stress-test and uncertainty controls affect the short-term planning check; they do not directly change investment results.

9. Risk Profile and Asset Universe

This module prepares the educational investment setup used by the Strategy Engine. The user chooses an investment philosophy, selects or customises an asset universe, and checks the cached market-data panel that will feed the engine.

Important distinction

No portfolio optimisation happens in this module. This screen chooses the risk profile, asset basket, and market-data panel. Optimisation, walk-forward testing, and performance evaluation happen later in the Strategy Engine.

9.1 Purpose of the Module

- Choose the broad investment philosophy used by later modules.
- Select the universe basket that will be available to the Strategy Engine.
- Allow a custom asset list when the user wants to override the generated basket.
- Confirm that the market-data panel is ready before running the Strategy Engine.
- Carry forward the funding bridge from the Personal Finance Planner.

9.2 Risk Profile Options

The user chooses an investment philosophy from three broad profiles. These profiles describe the educational posture of the scenario; they are not personal recommendations.

Risk profile	Meaning in the prototype
Growth	A more return-seeking posture that allows larger swings in pursuit of higher long-term growth.
Balanced	A middle-ground posture that aims to participate in upside while limiting extreme behaviour. In the example screenshots, this profile is associated with a typical return range of about 6%-9%, medium volatility, and a typical universe of 25-50 assets.
Defensive	A more cautious posture that prioritises lower volatility, stability, or capital preservation over aggressive growth.

9.3 Universe Basket Options

The universe basket defines which assets are available to the Strategy Engine. The default screen shows a recommended basket for the selected risk profile, but the user can reveal additional options.

- Show all sizes: reveals additional universe-size options such as 12, 25, 50, 75, and 100 assets. Smaller universes are easier to inspect; larger universes may provide broader coverage but are more complex.
- Show all strategies: reveals additional basket strategies beyond the recommended option.
- Universe size: sets the intended number of assets in the selected basket.
- Universe strategy: sets the type of asset basket used for the scenario.

- Use custom asset list: allows the user to override the generated preset basket with a manually selected asset list.

Available universe strategies shown in the interface include Core multi-asset, Diversified global beta, Equity heavy, Defensive income, Real assets tilt, Quality / Dividend equity, Low volatility / capital preservation, and Long-history / projection-friendly.

9.4 Custom Asset List

If Use custom asset list is enabled, the user can manually edit the selected assets. The interface shows the selected ticker chips and confirms that a custom universe override is active.

Turning the custom override off returns the module to the generated preset universe. The app keeps the previous custom selection so the user can turn it back on without rebuilding it from scratch.

Custom universe note

A custom basket changes the assets available to the educational engine, but it still does not create personalised financial advice or a guaranteed portfolio.

9.5 Universe and Data Details

The Universe and data details expander gives transparency about the selected basket. It can show the universe mix by broad asset group and the full asset list with ticker, name, group, and subgroup.

- Universe mix: summarises the selected assets across groups such as Equity, Fixed Income, Commodity, and Real Estate.
- Full asset list: shows the tickers and labels included in the selected basket.
- Subgroups: provide more specific labels such as US equities, Treasury bonds, gold, or sector equity.

In the example screenshots, the Balanced / Core multi-asset setup uses a 25-asset recommended basket. The universe mix shows 18 equity assets, 4 fixed-income assets, 2 commodity assets, and 1 real-estate asset.

9.6 Market-Data Setup

The market-data setup controls the panel that will be used by the Strategy Engine. In the hosted demo, the app uses cached Yahoo-generated panels by default to avoid live Yahoo/rate-limit issues and make the deployed version more reliable.

Field	Meaning
Panel start date / end date	Defines the historical coverage used to prepare the market-data panel. The screenshots show an example range from 2005/01/01 to 2026/05/03.
Return frequency	Sets the return frequency used by the prepared panel. Monthly is the recommended deployment frequency.
Yahoo auto-adjusted prices	Applies to live/local Yahoo downloads only. Cached deployment CSVs are already prepared.
Cached deployment panel	Indicates that the hosted app is using the prepared cached market-data source rather than downloading live data during the demo.

9.7 Sidebar Guidance

The sidebar provides a compact summary of the current setup while the user works in this module. It normally includes the current universe, the funding bridge inherited from the Personal Finance Planner, and the market-data status.

- Current universe: shows the selected risk profile, basket, selected size, and whether the module is using the recommended basket or a custom override.

- Funding bridge: shows the monthly contribution and weekly equivalent passed forward from the Personal Finance Planner.
- Market-data status: confirms whether the panel is ready for the Strategy Engine, including asset count, row count, return frequency, and data source.
- Risk & Universe Q&A: explains that this screen prepares assets and market data but does not optimise the portfolio.
- Risk & Universe terms: defines terms such as risk profile, universe basket, asset groups, market-data panel, and funding bridge.

In the example screenshots, the sidebar shows a Balanced risk profile, Core multi-asset basket, 25 selected assets, and a funding bridge of about £208/month or £48/week from the Personal Finance Planner. The market-data status shows that the cached monthly panel is ready for the Strategy Engine.

9.8 Diagnostics and Downloads

Diagnostics are optional and are hidden during the normal setup flow. The sidebar button can show or hide market-data diagnostics and downloads without refreshing data or running the engine.

- Prepared panel preview: shows a sample of the return/features panel that will feed the engine.
- Market-data timing breakdown: shows load and preparation timings, useful for transparency and debugging.
- Ticker audit: compares requested assets with assets loaded in the prepared panel and reports missing tickers.
- Downloads: provides CSV export controls for the prepared panel and raw cached panels where available.

The screenshots show an example cached panel with 37 requested assets, 37 loaded assets, and 0 missing. This diagnostic information is useful for auditability, but it is not required for the basic user flow.

9.9 Main Outputs

- Current setup summary: risk profile, number of selected assets, universe strategy, and whether the basket is recommended or custom.
- Universe mix and full asset list: transparent view of the selected asset basket.
- Market-data panel status: confirmation that the panel is ready for the Strategy Engine.
- Funding bridge: contribution value inherited from the Personal Finance Planner.
- Diagnostics and CSV exports: optional technical preview and audit information.

9.10 How to Interpret This Module

This module should be interpreted as a setup and data-preparation step. It decides what universe and data the later Strategy Engine can use, but it does not decide portfolio weights or evaluate performance.

The selected basket is the intended universe. The prepared panel may sometimes differ from the selected basket because it reflects the cached or available market data used by the engine.

The Continue to Strategy Engine button should be used once the current setup and market-data status look correct.

9.11 Caveats for This Module

- Risk profiles and universe baskets are educational setup options, not investment recommendations.
- No optimisation or backtest result is produced on this screen.
- Cached market data improves hosted-demo reliability but may not always match a live Yahoo download at a later date.
- Asset availability depends on the prepared market-data source.

- The funding bridge is inherited from the finance setup and remains an estimate rather than verified bank data.

10. Strategy Engine

The Strategy Engine turns the selected risk profile, asset universe, and market-data panel into a tested strategy return series. The engine creates the tested portfolio path first; headline metrics such as CAGR, volatility, Sharpe ratio, and maximum drawdown are calculated afterwards.

Important interpretation note

This is a historical strategy test for comparison inside the prototype. It is not a forecast, guarantee, or live fund track record.

10.1 Purpose of the Module

- Run the selected universe through a historical walk-forward strategy engine.
- Create a tested out-of-sample return path from the chosen setup.
- Calculate headline metrics after the portfolio path exists.
- Provide a result that can feed the Long-Term Scenario Explorer.
- Offer optional improvement checks and diagnostics for transparency.

10.2 Main Setup Options Before Running

Before the portfolio run, the user chooses a strategy template and a style preset. These are high-level controls intended for normal users; they avoid exposing all internal engine parameters by default.

Control	Available options shown	Purpose
Strategy template	Balanced Risk-Controlled, Core Ranking, Hybrid Research	Selects the broad engine method used to build the tested portfolio.
Style preset	Defensive, Balanced, Growth	Applies a risk posture to the selected template.
Fine-tune strategy settings	Optional posture sliders	Lets the user adjust broad behaviour without directly editing technical engine parameters.
Run portfolio & view results	Main action button	Runs the selected setup and produces the tested result.

The main pre-run screen also displays a green readiness message once the selected universe and market-data panel are loaded. Before the engine has been run, the Long-Term Scenario button remains unavailable because no tested strategy result exists yet.

10.3 Strategy Templates and Style Presets

The Strategy template changes the broad portfolio-construction method. The Style preset changes the risk posture applied to the selected template. These controls are user-facing setup choices, not financial recommendations.

- **Balanced Risk-Controlled:** a risk-aware setup designed to balance growth and drawdown control.
- **Core Ranking:** a more direct ranking-style setup that focuses on selecting assets through the engine ranking process.
- **Hybrid Research:** a more research-oriented setup that can be useful for experimentation and evaluation.
- **Defensive style:** more cautious posture, typically prioritising stability and lower drawdown.
- **Balanced style:** middle-ground posture between growth and stability.
- **Growth style:** more return-seeking posture, accepting more variability in exchange for potential upside.

10.4 Optional Suggestion Testing Controls

The advanced suggestion testing controls affect optional improvement checks after the portfolio run. They do not replace the main portfolio run. Higher testing depth checks more alternatives, but it can add noticeable runtime.

Testing depth	Meaning
Fast	Tests fewer candidates and is useful for quick checks.
Balanced	Default middle-ground depth. In the screenshots, it tests 1 preset, 2 tuning candidates, 4 universe-mix candidates, and 1 size test.
Thorough	Tests more alternatives and may take longer.
Custom	Allows more deliberate testing configuration when supported by the interface.

The interface warns that deeper tests add real engine runtime after the portfolio run. This is why the control is hidden unless the user deliberately opens it.

10.5 Fine-Tune Strategy Settings

The Fine-tune strategy settings expander gives non-technical posture controls. These sliders currently match the selected preset defaults unless the user changes them. The Reset preset sliders button returns them to the selected preset baseline.

Slider	What it changes conceptually
Risk appetite	How much return-seeking behaviour the setup allows.
Drawdown protection	How strongly the setup tries to limit painful declines.
Diversification <-> concentration	Whether the portfolio is broader or more concentrated.
Overlay intensity	How strongly additional risk-control or overlay behaviour is applied.
Stability <-> responsiveness	Whether the engine changes slowly or reacts more quickly.
Confidence in signal	How strongly the engine trusts its ranking or return signal.
Low turnover <-> adaptive	Whether the strategy prefers lower turnover or more frequent adaptation.
Simplicity <-> sophistication	How simple or technically layered the setup is intended to be.

For normal testing, these controls can be left at their preset values. They are included so the prototype can demonstrate adjustable behaviour without forcing users into low-level technical parameters.

10.6 Advanced Technical Engine Overrides

The technical overrides are optional and should usually be left unchanged. They are intended for deliberate technical evaluation of engine behaviour. If the user is unsure what a parameter means, the interface encourages checking the help icon before changing it.

Technical control	Plain-English meaning
top_k	How many assets are selected or emphasised by the engine.
lookback_mu	Lookback window used for return or expected-return estimates.
lookback_sigma	Lookback window used for risk or volatility estimates.
signal_mode	Type of return/risk signal used by the engine, such as mu_sigma.
temperature	How strongly the ranking translates into allocation preference.
inertia	How much the engine resists changing allocations too quickly.
weight_shrink	How much weights are pulled toward a smoother or less extreme allocation.
feature_mu_enabled	Whether feature-conditioned return tilts are enabled.
feature_mu_blend	How strongly feature-conditioned tilts affect the return estimate when enabled.

Technical controls caveat

These parameters are useful for auditability and experimentation, but they are not required for the basic user flow and should not be treated as investment advice.

10.7 Pre-Run Sidebar Guidance

Before running the engine, the sidebar summarises the current engine setup and shows whether the module is ready. In the screenshots, the sidebar shows a Balanced Risk-Controlled template, Balanced style, Balanced risk profile, 37 panel assets, 9,170 rows, and a funding bridge inherited from the Personal Finance Planner.

- Engine setup: template, style, risk profile, and universe/data size.
- Funding bridge: monthly contribution and weekly equivalent from the Personal Finance Planner.
- Run status: before the run, the sidebar shows that the Strategy Engine has not run yet.
- Strategy Engine Q&A: explains why the engine is needed and what it can improve or worsen.
- Strategy Engine terms: defines terms such as strategy template, style preset, out-of-sample path, CAGR, volatility, Sharpe, and max drawdown.

10.8 Running the Portfolio

The main action is Run portfolio & view results. Once selected, the app runs the historical strategy test using the selected engine setup, asset universe, market-data panel, and funding context. The user should wait for the green completion message before interacting with post-run outputs.

After the run completes, the screen changes to Strategy Engine Results. A green status message confirms that results are ready for review and long-term scenario projection.

10.9 Headline Result Metrics

The Strategy Engine presents the main historical backtest metrics at the top of the results screen. In the example screenshots, the current run produced:

Example Strategy Engine result

CAGR: 9.73% | Volatility: 11.71% | Max drawdown: -19.43% | Sharpe: 0.85

Metric	How to read it
CAGR	Annualised growth rate of the tested strategy over the evaluated historical window. It is not a guaranteed future return.
Volatility	How bumpy or variable the tested return path was. Higher volatility usually means the journey felt less stable.
Max drawdown	Worst historical fall from a previous high to a later low. It is the main pain-test metric.
Sharpe	Risk-adjusted return measure. It compares return with volatility, but it depends on the tested period and assumptions.

The How to read this result expander interprets the metrics relative to the selected risk profile rather than as universal ratings. For a Balanced profile, the example run is described as broadly balanced, with positive long-run growth and a usable risk-adjusted profile, while the drawdown still matters.

10.10 Benchmark Context

The results panel can show same-period context against simple references such as the S&P 500, Nasdaq-100, and Gold. This helps the user understand whether the tested strategy behaved more like a growth benchmark, a defensive diversifier, or a risk-controlled profile.

In the example screenshots, the tested strategy had lower CAGR than the S&P 500 and Nasdaq-100, but also lower volatility and a less severe maximum drawdown than Nasdaq-100. This supports a risk-controlled interpretation rather than a pure growth-benchmark interpretation.

10.11 Optional Improvement Checks

After a run, the app can show optional improvement checks. These checks are not required to use the system, but they help the user compare controlled alternatives before moving into long-term projection. They are organised as phases so the user can decide whether to apply a suggested change or keep the current tested setup.

Phase	Focus	User decision
Phase 1 - Strategy preset	Tests an alternative template/style combination.	Apply the recommended preset or keep the current preset and continue.
Phase 2 - Engine tuning	Tests small technical parameter changes within the current strategy family.	Apply the recommended tuning or keep the current tuning and continue.
Phase 3 - Universe mix	Tests same-size changes to the asset mix.	Apply the recommended universe composition or keep the current mix.
Phase 4 - Universe size	Tests whether a broader or narrower basket size improves the trade-off.	Apply a size change if it passes the gate, or keep the current size and finish.

Each phase can appear as Active, Applied, Locked, or Skipped. Active means the user can decide on that phase now. Applied means a recommendation was accepted. Locked means the phase is not available until earlier phases are resolved. Skipped means the user kept the current setup, or the app did not find a tested alternative that passed the acceptance gate.

Decision guide

Apply a suggestion only if the rerun-tested trade-off feels better for the selected risk profile. A single improved metric is not enough if the overall CAGR / volatility / drawdown / Sharpe balance becomes worse.

The checks preserve user control. The app can recommend a change, but the user chooses whether to apply it or continue with the current setup. This prevents the suggestions from being treated as automatic advice.

10.12 Example Recommendation Flow

The screenshots show a representative improvement-check flow. The initial Strategy Engine result produced CAGR 9.73%, volatility 11.71%, maximum drawdown -19.43%, and Sharpe 0.85.

Stage	Example outcome	How to read it
Baseline run	CAGR 9.73%, Vol 11.71%, MaxDD -19.43%, Sharpe 0.85	The starting historical strategy result for comparison.
Phase 1 - Strategy preset	Balanced Risk-Controlled + Defensive; CAGR 9.93%, Vol 11.61%, MaxDD -18.97%, Sharpe 0.88	The preset candidate improved the displayed trade-off and passed the acceptance gate.
Phase 2 - Engine tuning	Balanced middle-ground tuning; CAGR 10.00%, Vol 11.57%, MaxDD -18.83%, Sharpe 0.89	The tuning candidate keeps the same broad strategy and adjusts small technical knobs.
Phase 3 - Universe mix	Diversified available-pool composition applied	The asset mix can be changed while keeping the same broad size.
Phase 4 - Universe size	Current universe size kept; tested size alternative did not materially improve the result	If no size alternative passes the gate, the app keeps the current basket size.

The final improvement flow can therefore show several green Applied phase cards and a yellow Skipped phase card. This does not mean the app failed; it means the current setup was kept where no tested alternative improved the overall trade-off enough.

10.13 Apply, Keep, and Skip Behaviour

- Apply recommended preset / tuning / universe change: accepts the rerun-tested candidate and carries it forward into later phases.
- Keep current preset / tuning / size and continue: rejects or skips the current suggestion while keeping the current tested result or setup.
- Passed gate: the app found that a candidate improved the overall decision trade-off enough to be recommended.
- No accepted candidate: the app keeps the current setup because tested alternatives did not improve enough to pass the acceptance gate.

- Historical suggestion tests are not re-run after Apply; the app records the tested suggestion path and uses it for the improvement flow.

The important user behaviour is to compare the whole profile of the result, not only one number. For example, a slightly higher CAGR may not be attractive if it creates a much larger drawdown or a less stable strategy.

10.14 Run Timings and Diagnostics

Run timings and diagnostics can be opened from the sidebar after a Strategy Engine run. They are technical support information rather than normal user-facing recommendations. They help explain why some runs take longer online and how much time was spent in different engine components.

- Engine timing: shows overall engine runtime and major components such as the walk-forward loop and probabilistic layer.
- Suggestion timing: shows how long optional improvement checks took and how many candidates were tested or passed.
- Detailed timing tables: show component-level timings and candidate-level suggestion timings when expanded.
- Technical metadata: records the source, panel shape, run signature, config fingerprint, and timestamp for auditability.

In the example screenshots, the initial engine run takes around 43-45 seconds, while optional suggestion testing can add several minutes depending on the number of candidates and phases tested. Timing varies by deployed environment, cache state, panel size, and selected testing depth.

10.15 Reliability and Robustness Assessment

Reliability and robustness should normally be reviewed after optional improvement checks are completed, because accepted suggestions can change the setup being assessed. The reliability panel helps the user understand whether the result is supported by the available historical evidence and whether it is sensitive to the chosen historical start window.

Element	Meaning
Overall reliability	A plain-English confidence label based on available market data, walk-forward evidence, benchmark checks, and method caveats.
Start-date robustness	A sensitivity check that reruns the same configuration from different historical start dates within the existing panel.
Detailed evidence and validation	Audit-style evidence showing reliability components, benchmark validation, and method notes.
Benchmark calculation validation	Compares benchmark metrics shown by the app against an independent recomputation from the same selected market-data panel and evaluated period.
Method notes and limits	Explains that results are useful for comparing configurations inside the app, but are not forecasts or guarantees.

In the example screenshots, the reliability assessment reports Overall reliability: Moderate-to-Strong and Start-date robustness: Strong. The robustness check tested multiple historical start windows and reported a CAGR range of about 8.56% to 11.26%, a Sharpe range of about 0.75 to 0.99, and a MaxDD range of about -15.77% to -18.92%.

Reliability interpretation

A stronger reliability label means the result looked more stable across the checks performed inside the app. It does not make the result predictive and does not remove the usual limitations of historical backtesting.

10.16 Start-Date Robustness Details

The start-date robustness check reruns the same Strategy Engine configuration using different historical start dates from the existing selected market-data panel. It checks whether the result depends too heavily on one specific historical window.

Start date	Example status / interpretation
2005-01	Baseline reused; current baseline window.
2008-01	Executed rerun; robustness candidate.
2010-01	Executed rerun; robustness candidate.
2012-01	Executed rerun; shorter but usable window.
2015-01	Too short / diagnostic in the example screenshots.

The screenshots show that five start dates were checked, three reruns were executed, and the robustness test took about 74 seconds. Shorter windows should inform interpretation rather than automatically invalidate the result.

10.17 Technical Improvement Guide

The Technical improvement guide is included for deliberate experimentation. Before a run, it explains possible test hypotheses; after a run, it can adapt to the latest CAGR, volatility, MaxDD, Sharpe, and philosophy. It indicates which controls are most likely to affect growth, volatility, drawdown, or Sharpe.

These are tests, not guarantees. The engine only recommends a change if the rerun-tested trade-off passes the relevant acceptance gate.

10.18 Main Outputs

- Tested strategy return path and headline metrics.
- CAGR, volatility, maximum drawdown, and Sharpe ratio.
- Plain-English metric interpretation for the selected risk profile.
- Same-period benchmark context.
- Optional improvement checks and recommendation decision guide.
- Reliability and start-date robustness assessment.
- Run timings, candidate timings, and technical metadata.
- Stored result that can be used by the Long-Term Scenario Explorer.

10.19 How to Interpret This Module

The Strategy Engine should be interpreted as the historical test layer of the prototype. It shows how one selected setup behaved over the available tested period, including risk and drawdown, rather than promising how it will behave in the future.

A good result is not just the highest CAGR. The user should read CAGR together with volatility, maximum drawdown, Sharpe, benchmark context, improvement-check decisions, reliability, and start-date robustness.

The most important practical output is the stored tested result. Once the run completes and the user is satisfied with the selected setup, the user can continue to the Long-Term Scenario Explorer with an actual tested strategy path rather than only a generic educational proxy.

10.20 Caveats for This Module

- Historical backtest results are not forecasts or guarantees.
- The engine can improve one metric while worsening another, so the whole trade-off should be considered.
- Optional improvement checks add runtime and are still educational tests, not advice.
- Applied suggestions are controlled scenario choices, not personalised recommendations.
- Advanced technical overrides can make results harder to interpret if changed without a clear hypothesis.
- Reliability and timing diagnostics are useful for transparency, but they do not make the result predictive.
- Start-date robustness checks sensitivity to historical windows; it does not predict future regimes.

11. Long-Term Scenario Explorer

The Long-Term Scenario Explorer translates the contribution bridge and return-path assumption into possible long-term wealth ranges. It compares a savings-only baseline with either a completed Strategy Engine return path or a clearly labelled educational investment proxy.

Important interpretation note

Long-Term Scenario is a scenario explorer, not a forecast. It does not predict an exact future amount. It shows possible low / middle / high outcome ranges under the selected assumptions.

11.1 Purpose of the Module

- Translate the monthly contribution from Personal Finance Planner into long-term scenario outcomes.
- Use the completed Strategy Engine return series when available.
- Compare an investment/proxy path with a savings-only baseline.
- Show uncertainty through P10, median, and P90 ranges rather than one promised number.
- Provide scenario outputs that can be summarised in the final report.

11.2 Projection Inputs and Settings

The module can run from the current contribution plan and the latest tested Strategy Engine path. In the screenshots, the scenario uses a contribution of about £208/month, equivalent to about £48/week, and a Balanced projection profile.

Setting	Meaning
Starting investment pot	Amount already available at the start of the long-term scenario. It is separate from future monthly contributions.
Optional wealth goal	Target amount used to calculate goal probability. If left at 0, the scenario can still run, but goal probability is less meaningful.
Compare across horizons	Runs the same contribution and return-path assumptions across selected time horizons such as 20, 30, and 50 years.
Show savings-only baseline	Adds a cash/savings-only comparison path that uses the same starting pot and contribution but no investment return, volatility, drawdown, or market risk.
Horizon list	Controls which long-term horizons appear in the comparison table and chart.
Advanced Monte Carlo/path settings	Optional controls for the number of paths, path mode, synthetic trading days, and within-month path variation.

11.3 Advanced Monte Carlo / Path Settings

The advanced path settings are optional. They allow the user to inspect or adjust how the scenario range is generated. Normal users can usually keep the suggested settings.

Control	Plain-English meaning
Monte Carlo paths	Number of simulated paths used to build the scenario range. More paths can make the range smoother, but they do not make the future more predictable.
Projection path mode	Controls whether the scenario uses monthly returns directly or expands each sampled month into a synthetic within-month journey.
Hybrid daily simulation	Samples monthly Strategy Engine returns, then expands each sampled month into synthetic daily steps that compound back to the same monthly return.
Synthetic trading days per month	Number of artificial daily steps used inside each simulated month when hybrid daily simulation is selected.
Daily path variation	Controls how much visible movement appears inside each simulated month. It changes the shape of the path, not the sampled monthly return itself.

In the screenshots, the app suggests a Balanced projection coherence setting: hybrid daily simulation, 21 synthetic trading days per month, and 0.18 daily path variation. This is described as realistic but not overly aggressive for the selected profile.

11.4 Sidebar Guidance

The sidebar explains the scenario context and keeps the main inputs visible while the user reviews the projection. It normally includes the scenario branch, contribution, weekly equivalent, projection profile, return path, latest engine run, and whether the projection has been generated on the current screen.

- Scenario status: summarises the contribution plan and whether the scenario is using a tested Strategy Engine path or an educational proxy.
- Projection context: shows market-data panel information, the last Strategy Engine metrics, and how many monthly out-of-sample returns are available for the projection bridge.
- Scenario Q&A: explains what the screen is for, why it is useful, and how it differs from the Strategy Engine.
- Scenario terms: defines starting pot, wealth goal, horizon comparison, P10 / median / P90, Monte Carlo paths, projection path mode, savings-only baseline, educational proxy, loss vs contributions, and goal probability.

11.5 Savings-Only vs Investment / Proxy Comparison

The savings-only baseline is the stability comparison. It uses the same starting pot and monthly contribution but applies no investment return, volatility, drawdown, or market risk. The investment/proxy path is not presented as a guaranteed better choice. It shows how a disciplined long-term strategy could change the range of outcomes if the user can tolerate uncertainty, temporary losses, and model risk.

How to use the comparison

Use this module to understand trade-offs: cash-flow stability first, then long-term scenario testing. The app does not recommend investing and does not predict future returns.

11.6 Main Outputs

The projection output presents a comparison table and summary metrics. In the example screenshots, the projection compares the current one-year view with 20, 30, and 50-year horizons.

Output	Meaning
Horizon comparison table	Compares terminal outcomes at each selected horizon, including median final wealth, savings-only baseline, P10, P90, and loss vs contributions.
Median final wealth	Middle outcome across simulated paths for the selected horizon.
P10 / P90	Lower and higher scenario range markers. They are not promised outcomes.
Savings-only baseline	Amount from contributions and starting pot only, without investment return or market volatility.
Loss vs contributions	Share of simulations where the terminal value ends below total contributions. It is a scenario diagnostic, not a prediction.
Goal probability	Share of simulations that reach the optional wealth goal. It should be treated as a secondary diagnostic unless the goal is meaningful.

11.7 Example Projection Values

The screenshots show an example horizon comparison with the current one-year projection and longer horizons. The one-year result has a median final wealth of about £2,660, a savings-only baseline of £2,496, a P10 value of about £2,438, a P90 value of about £2,846, and a loss-vs-contributions diagnostic of about 22%.

Horizon	Median final wealth	Savings-only baseline	P10	P90
1 year (current)	£2,660	£2,496	£2,438	£2,846
20 years	£152,118	£49,920	£103,207	£253,314
30 years	£450,517	£74,880	£247,456	£853,247
50 years	£3,316,735	£124,800	£1,610,992	£7,301,425

These figures are included as example scenario outputs from the screenshots. They depend on the selected contribution, engine path, horizon, simulation settings, and data period. They should not be interpreted as expected real-world future wealth.

11.8 How to Interpret This Module

The Long-Term Scenario Explorer asks a different question from the Strategy Engine. The Strategy Engine asks how a selected strategy behaved on the historical market-data panel. The Long-Term Scenario asks what range of contribution outcomes could occur if future returns behaved similarly to the selected return path or proxy.

The user should focus on ranges and trade-offs rather than one headline number. A high median can still come with uncertainty, downside outcomes, and periods where the path may feel uncomfortable.

The module is most useful after the Strategy Engine has produced a tested path, because the scenario can then use the historical Strategy Engine out-of-sample return series instead of a generic educational proxy.

11.9 Caveats for This Module

- Monte Carlo outputs are scenario simulations, not predictions.
- The projection range depends on the selected contribution, return path, data window, simulation settings, and modelling assumptions.
- The savings-only baseline is not risk-free in real life because inflation and purchasing-power risk are not fully modelled.
- The investment/proxy path is not a recommendation to invest and is not guaranteed to outperform savings-only.
- More Monte Carlo paths can make the displayed range smoother, but do not make future outcomes more certain.
- The module works at strategy-return level and does not yet show month-by-month asset-level purchase percentages.
- Essential spending is treated as fixed in the model; discretionary contribution behaviour can be adjusted by the prototype.

12. Final Report and Insights Summary

The Final Report is the closing screen of the recommended workflow. It summarises the selected personal finance plan, investment setup, Strategy Engine result, Long-Term Scenario comparison, and decision-support interpretation in one place.

Evidence level

In the example screenshots, the report uses a tested Strategy Engine result interpreted through the Long-Term Scenario. This is stronger than a generic educational proxy, but it is still historical scenario evidence rather than a forecast.

12.1 Purpose of the Final Report

- Show how the personal finance setup, investment setup, Strategy Engine result, and long-term projection connect.
- Summarise the main planning comparison without requiring the user to revisit every previous screen.
- Explain the trade-off between savings-only simplicity and investment/proxy upside uncertainty.
- Provide a final decision-support view for evaluation, not a recommendation or guarantee.
- Confirm that the full end-to-end prototype flow has produced a coherent report.

12.2 Executive Summary

The executive summary highlights the main cross-module values. In the example screenshots, the report shows a monthly contribution of about £208/month, a 50-year savings-only baseline of £124,800, a 50-year investment/proxy median of £3,316,735, and a 50-year modelled difference of £3,191,935.

Bottom line

At 50 years, the investment/proxy path has a higher modelled median outcome than savings-only in the example. That upside only matters if the uncertainty, drawdown exposure, model assumptions, and user tolerance are acceptable.

Executive metric	Example value	Meaning
Monthly contribution	£208/month	Contribution bridge inherited from the Personal Finance Planner.
50-year savings-only	£124,800	Same contribution path without investment return or market volatility.
50-year investment/proxy median	£3,316,735	Median modelled outcome under the selected return-path assumptions.
50-year difference	£3,191,935	Difference between the investment/proxy median and savings-only baseline.

12.3 Strategy and Setup Snapshot

The report also summarises the tested strategy and setup used by the scenario. In the screenshots, the Strategy Engine snapshot shows Sharpe 0.99, CAGR 10.2%, and drawdown severity 18.9%.

Snapshot area	What it shows
Personal finance plan	Contribution capacity used in the scenario, such as £208/month.
Investment setup	Return path source, risk profile, and whether the Strategy Engine has been tested.
Strategy Engine snapshot	Key historical metrics such as Sharpe, CAGR, and drawdown severity.
Long-Term Scenario	Which horizons were compared, such as current 1 year, 20 years, 30 years, and 50 years.

12.4 Final Comparison Readout

The final comparison readout explains whether the investment/proxy pathway produced a higher median outcome than savings-only across the tested horizons. It does not say that investing is automatically better; it frames the result as a trade-off between upside, uncertainty, model risk, and drawdown exposure.

Horizon	Savings-only	Investing median	Investing P10	Investing P90	Difference
1 year (current)	£2,496	£2,660	£2,438	£2,846	£164
20 years	£49,920	£152,118	£103,207	£253,314	£102,198
30 years	£74,880	£450,517	£247,456	£853,247	£375,637
50 years	£124,800	£3,316,735	£1,610,992	£7,301,425	£3,191,935

12.5 Decision Interpretation

The decision interpretation section turns the numbers into plain-language planning guidance. It separates three ideas: upside versus simplicity, risk trade-off, and decision status.

Block	Meaning
Upside vs simplicity	Savings-only is simpler and avoids market volatility, but has limited upside. The investment/proxy path may improve the central outcome but introduces uncertainty.
Risk trade-off	A higher central scenario is not automatically better; it must be judged against drawdown exposure, model risk, evidence quality, and whether the user could stay invested through bad periods.
Decision status	The report encourages continuing to test assumptions rather than treating the result as a final answer.
Recommended next step	Stress-test the Long-Term Scenario with different contributions, goals, and horizons. Return to Strategy Engine only if the investment branch needs lower drawdown, a different universe, or a cleaner risk profile.

12.6 Selected-Horizon Details

The Selected-horizon projection details expander keeps the main report focused while still showing the detailed values for the currently selected horizon. In the screenshots, the selected-horizon values include expected

terminal wealth £2,652, median terminal wealth £2,660, P10 downside case £2,438, P90 upside case £2,846, and modelled growth £156.

Selected-horizon metric	Example value	Meaning
Expected terminal wealth	£2,652	Average-style model output for the selected horizon.
Median terminal wealth	£2,660	Middle scenario outcome.
P10 downside case	£2,438	Lower scenario range marker.
P90 upside case	£2,846	Higher scenario range marker.
Modelled growth	£156	Modelled increase above the contribution baseline for the selected horizon.

12.7 Sidebar Guidance

The final-report sidebar confirms that the report is available and lists the main state inputs used to generate it. It also provides report inputs, technical diagnostics, and report Q&A / terms expanders.

- Current report: confirms that the final report is available and summarises the selected plan, return path, and Long-Term Scenario comparison.
- Report inputs: checks that personal finance, market-data panel, strategy/proxy path, Long-Term Scenario, and final report are ready.
- Technical diagnostics: shows projection source, selected market-data panel, latest Strategy Engine metrics, and technical IDs such as run signature and config fingerprint.
- Report Q&A and terms: explains the report purpose, savings-only baseline, educational proxy, tested Strategy Engine path, scenario ranges, goal probability, and loss vs contributions.

Sidebar diagnostic caveat

The final-report sidebar reads app state only. It does not rerun projections, refresh data, or launch diagnostics.

12.8 How to Use the Final Report Safely

The report includes a dedicated safe-use section. This reinforces that the page should be used to compare pathways, assumptions, and trade-offs, not as financial advice or as a promise of future results.

- The Strategy Engine created a historical return path from the selected universe and engine configuration.
- The Long-Term Scenario translates contribution assumptions into projected ranges; it does not predict future wealth.
- P10, median, and P90 are scenario percentiles under the model assumptions, not guaranteed outcomes.
- The comparison is fair only when both branches use the same contribution, starting pot, goal, and horizons.
- Historical returns and educational proxies are not reliable indicators of future results.

12.9 Main Outputs

- Executive summary of contribution, savings-only baseline, investment/proxy median, and difference.
- Bottom-line explanation of why the planning comparison matters.
- Strategy Engine snapshot and setup snapshot.
- Final horizon comparison table with savings-only, median, P10, P90, and difference values.
- Decision interpretation covering upside, simplicity, risk trade-off, and next steps.
- Selected-horizon details for deeper inspection.
- Safe-use checklist and past-performance warning.
- Sidebar report inputs and technical diagnostics.

12.10 Caveats for This Module

- The final report summarises previous module outputs; it does not independently validate the user's real financial situation.
- A higher modelled median is not automatically a better decision because drawdown, uncertainty, model risk, and risk tolerance matter.

- The report is only as reliable as the assumptions, market-data panel, Strategy Engine result, and Long-Term Scenario settings used to create it.
- The report should support evaluation and reflection, not replace professional financial advice.
- The safest interpretation is comparative: how do different pathways behave under the same contribution and horizon assumptions?

13. Known Limitations and Future Work

LifeBudget Micro is a working educational prototype, but it is not a production financial planning platform. The submitted version demonstrates the core end-to-end flow while leaving several production and research extensions outside the current scope.

13.1 Current Limitations

- LifeBudget Micro is an educational prototype, not financial advice, investment advice, or a personal suitability assessment.
- The app is a historical what-if and scenario-comparison prototype. It shows how selected strategies would have behaved over available historical data; it does not predict future market outcomes.
- Investment outputs are based on historical backtests, educational proxies, and scenario assumptions. Historical performance does not guarantee future performance.
- The prototype does not fully model tax, ISA or pension rules, platform fees, fund charges, bid-ask spreads, inflation, currency effects, debt, liquidity needs, asset availability, or individual risk tolerance.
- The hosted Streamlit version uses prepared or cached market-data panels to keep the online deployment reliable. Market data availability may vary depending on the source and data preparation pipeline.
- The hosted version may be slower than local execution because market-panel preparation, backtesting, suggestion checks, and robustness diagnostics run on hosted Streamlit resources.
- The long-term projection works at strategy-return level and does not yet show month-by-month asset-level contribution allocation.
- Mobile browser access may work for quick inspection, but the prototype is primarily tested and optimised for desktop or laptop use.
- The system focuses on demonstrating core project logic and behaviour rather than production-grade financial planning, security, persistence, or large-scale deployment.

13.2 Not Implemented Yet / Future Development

- A production frontend could be rebuilt with a framework such as React, HTML, CSS, and JavaScript to improve responsiveness across desktop and mobile devices.
- A production data layer should replace cached CSV panels with robust API-based market-data pipelines, multiple data sources, fallback handling, and validation checks.
- Persistent user accounts and a database layer could allow users to save, reload, compare, and export their own plans, strategy setups, and scenario results.
- The current strategy controls could be expanded with richer parameter governance while keeping the default user interface simple and understandable.
- Additional governance layers, such as coherence checks, stability checks, and guided apply/reset controls, could be extended further in a future version.
- Recommendation diversity and scalability safeguards could be added to reduce convergence risk, where many users are pushed toward overly similar portfolio setups.
- The current historical what-if model could be extended into a forward-looking research mode using scenario variables such as inflation assumptions, interest-rate environments, central-bank policy regimes, and geopolitical stress scenarios.
- The Long-Term Scenario engine could be improved with more realistic contribution allocation, rebalancing, fees, inflation, alternative regimes, and asset-level contribution modelling.
- Future work could include multiple functional datasets from different providers rather than relying on one prepared cached market-data source for the hosted demo.

14. Troubleshooting

- If the app appears slow during market panel generation or strategy backtesting, wait for the relevant confirmation message before clicking additional buttons.
- If market data preparation fails temporarily, rerun the market panel generation step.
- If running locally, ensure that dependencies have been installed from the requirements file.
- If the hosted version becomes unresponsive, refresh the page and rerun the relevant section.
- If a chart or table is difficult to read on mobile, switch to a desktop or laptop browser.

15. Demo Recording Guide

1. Briefly introduce LifeBudget Micro and the educational notice.
2. Show the home screen and accept the educational assumptions.
3. Open Personal Finance Planner, set sample values, and show the feasibility result.
4. Prepare the risk profile and asset universe.
5. Run the Strategy Engine and show historical metrics and reliability context.
6. Open the Long-Term Scenario Explorer and generate a projection.
7. Finish with the final report and known limitations.

The demo should focus on showing that the system runs and that at least one meaningful feature works end-to-end.